CS & IT ENGINEERING



Algorithms

Heap Algorithms



Lecture No.- 02

By- Aditya sir

Recap of Previous Lecture



Topic

Topic

Intro to Heaps

→ Min Heap

→ Max Heap

→ Creation. → Insertion

→ Heapify

Topics to be Covered



- Topic
- Topic
- Topic

Operations

→ Heapify
→ Deletion
→ Heap Sort
→ PYQs



About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on Linkedin where I share my insights and guide students and professionals.



Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW

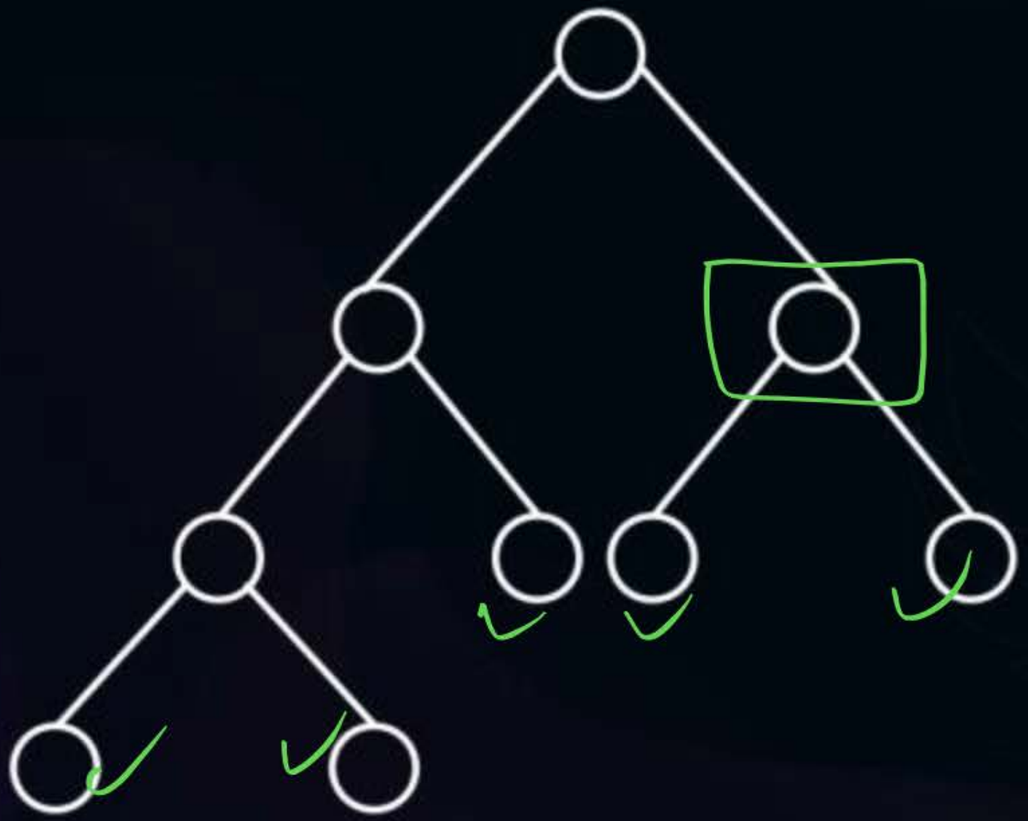


Topic : Heap Algorithm

Adjust Method:-

Heapify

Top-down



2nd Last level

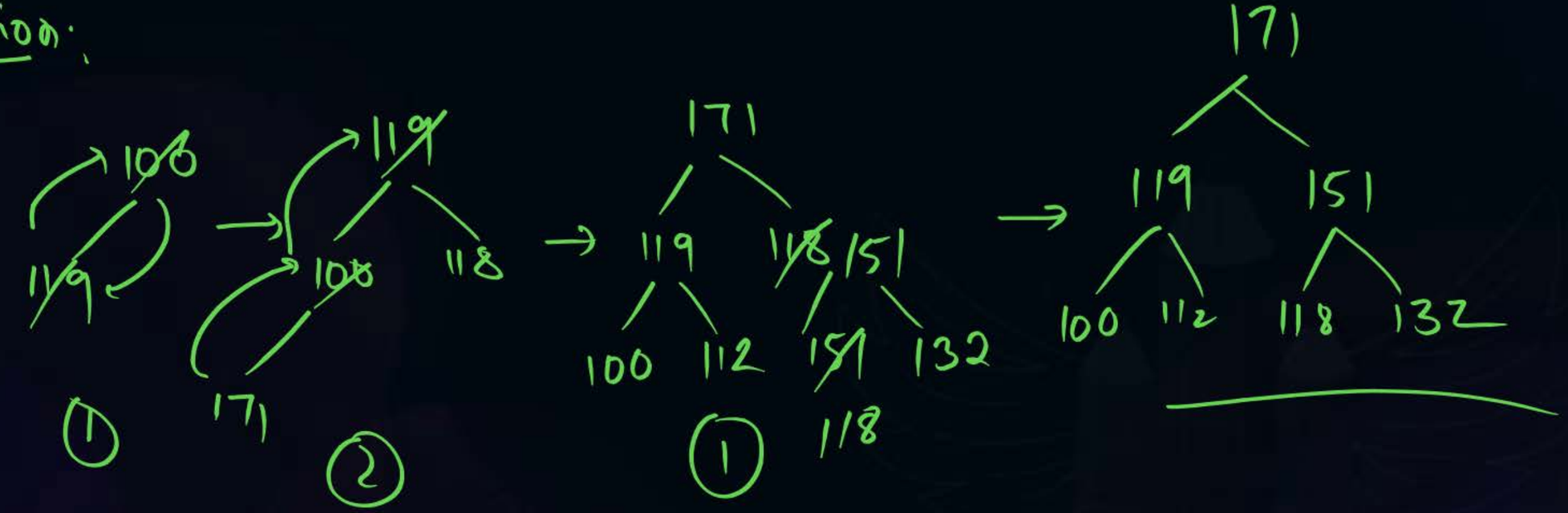
Topic : Heap Algorithm

E.g.1. A: [100, 119, 118, 171, 112, 151, 132]

68%

① Insertion:

② Heapify



① ② ① ④

Total swaps = 1+2+1 = 4



Topic : Heap Algorithm

#Q. How many swaps for heap creation using

(i) insertion method

(ii) Heapify / Build heap?



Topic : Heap Algorithm

48%

2. Heapify / Build Heap:

[100, 119, 118, 171, 112, 151, 132]



118 < 151

119 < 171

100 < 171

100 < 119

4 Swaps

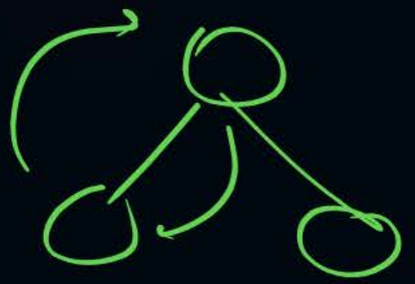




Topic : Heap Algorithm

Bonus

```
procedure ADJUST(A, i, n)
integer i, j, n;
j =2 * i; item ← A(i)
while i ≤ n do
        if (j ≤ n A (j) < A(j + 1)) then
        j ← j + 1 //j points to the larger child//
end if
if (item ≥ A(j)) then
        exit // a position for item is found //
else
        A(⌊j/2⌋) ← A(j)// move the larger child up a level//
j ← 2* j
end if
repeat A(⌊j/2⌋) ← item
end ADJUST
```





Topic : Heap Algorithm

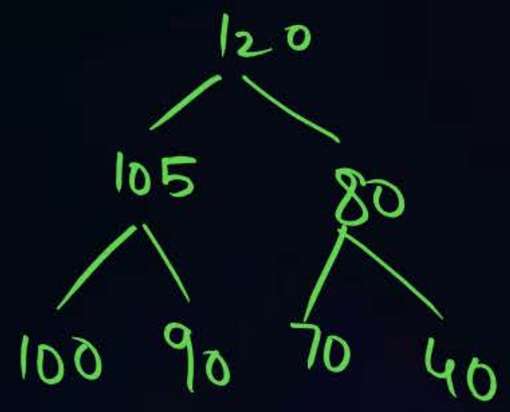
```
procedure HEAPIFY (A, n)
// Readjust the elements in A(1 : n) to form a heap//
Integer n, I
for i = ⌊n/2⌋ to 1 by – 1 do
        call ADJUST (A, i , n)
repeat
end HEAPIFY
```

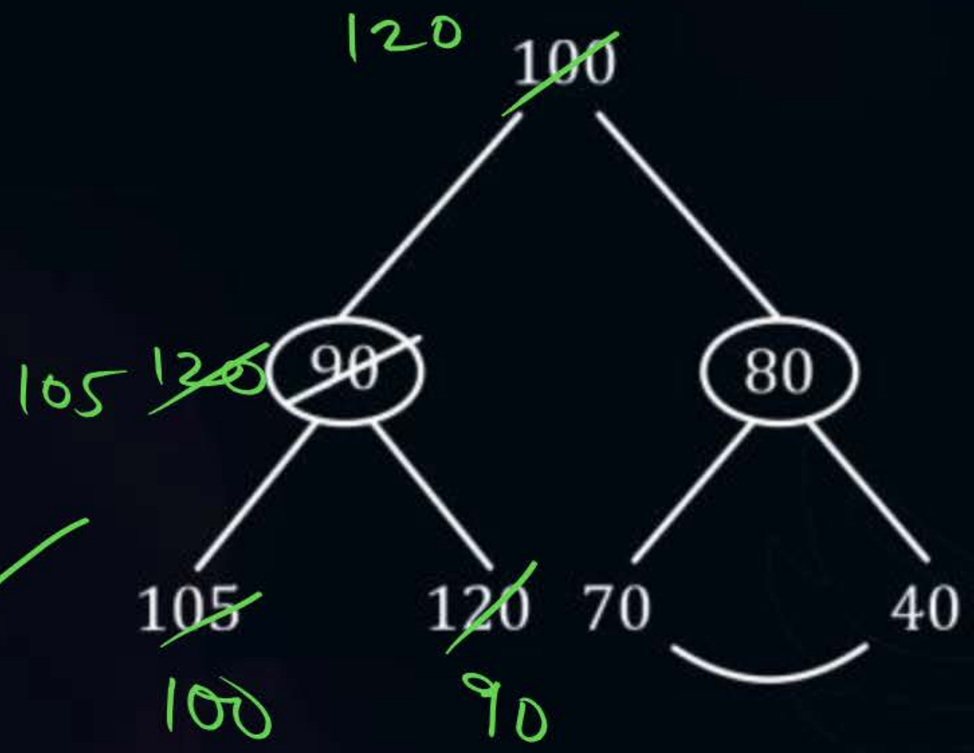
$\lfloor n/2 \rfloor \longrightarrow 1$





Topic : Heap Algorithm





$80 > \max(70, 40)$

$80 \geq 70$

$90 < 120$



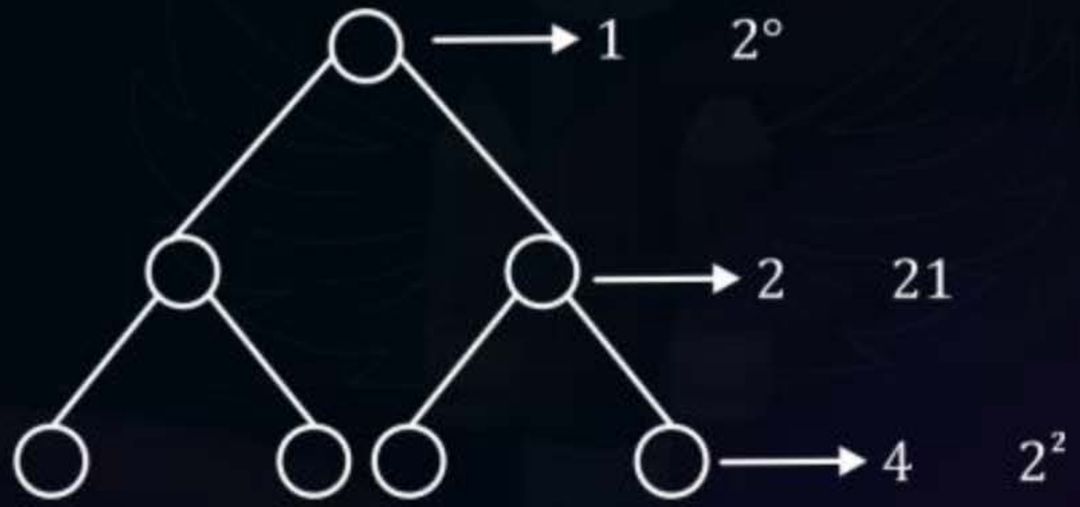
Topic : Heap Algorithm

Time complexity of Build-Heap (Heapify)

(logical approach) → (worst case)

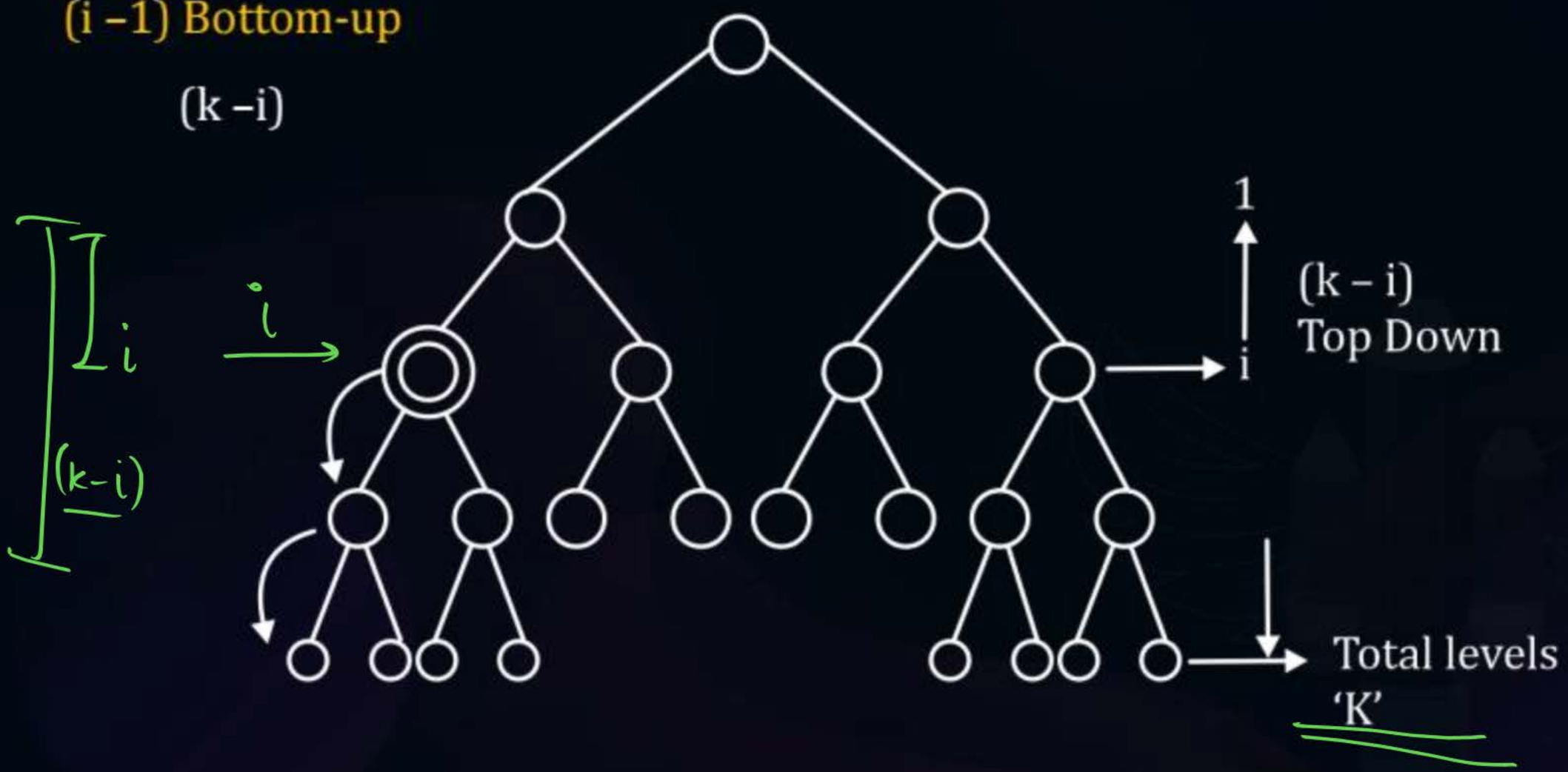
- Max number of nodes at nay level I = $2^{(i-1)}$ (root is at level 1)
- Max number of level comparisons / moves for a nodes present at any level I that is getting adjusted.

= (k –i)





Topic : Heap Algorithm





Topic : Heap Algorithm

3. For all the nodes at any level I, the total number of comparisons = $(k - i) * 2^{(i-1)}$
4. Total number of comparisons in worst case for all the nodes combines (level 1→ k)

$$T(n) = \sum_{i=1}^{k} (k - i) * 2^{(i-1)}$$

$$= \sum_{i=1}^{k} (k - i) * \frac{2^i}{2}$$

$$= \frac{1}{2}\left[\sum_{i=1}^{k} (k - i) \times 2^i\right]$$

$$= \frac{1}{2}\left[\sum_{i=1}^{k} k * 2^i - \sum_{i=1}^{k} i \times 2^i\right]$$



Topic : Heap Algorithm

$$= \frac{1}{2}\left[k(2^{k+1} - 2) - \left((k-1) * 2^{k+1} + 2\right)\right]$$

$$= \frac{1}{2}[2 \times k \times 2^k - 2 \times k - 2 - k \times 2^k \times 2 + 2^k \times 2]$$

$$= 2^k - k - 1$$

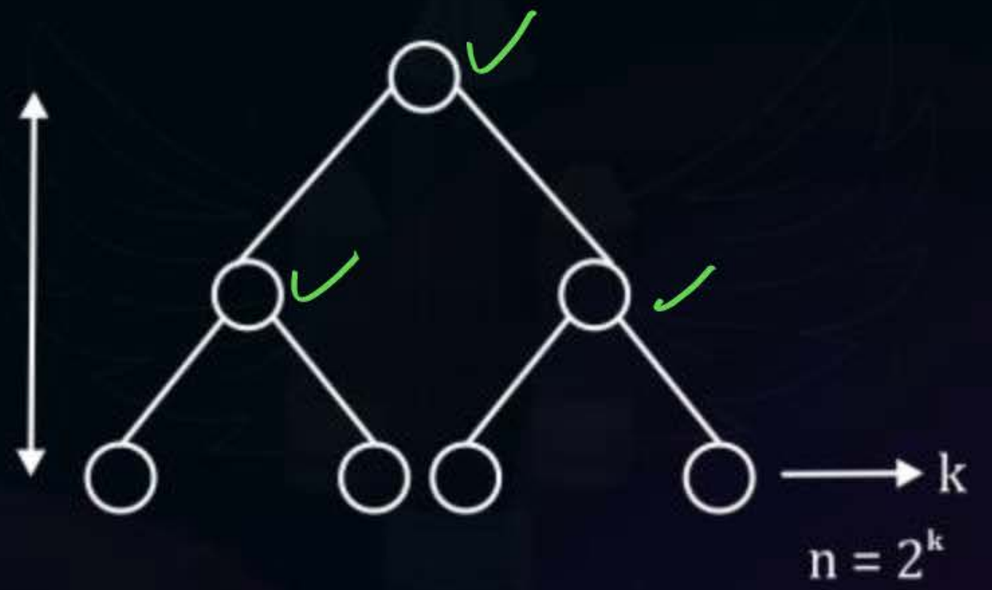
$= O(2^k)$

$= \Theta(n)$

$2^k \approx n$

$k = \log(n)$

$K = \log_2 n$





Topic : Heap Algorithm

$T(n) = 2^k - k - 1$

$= n - \log_2 n - 11$

$T(n) = O(n)$

To construct heap with n elements.

Heapify → O(n)

Insertion→ O(nlogn)

(CBT→ Readjust)

[redacted]: adding a node one-by-one to a Heap.



Topic : Heap Algorithm

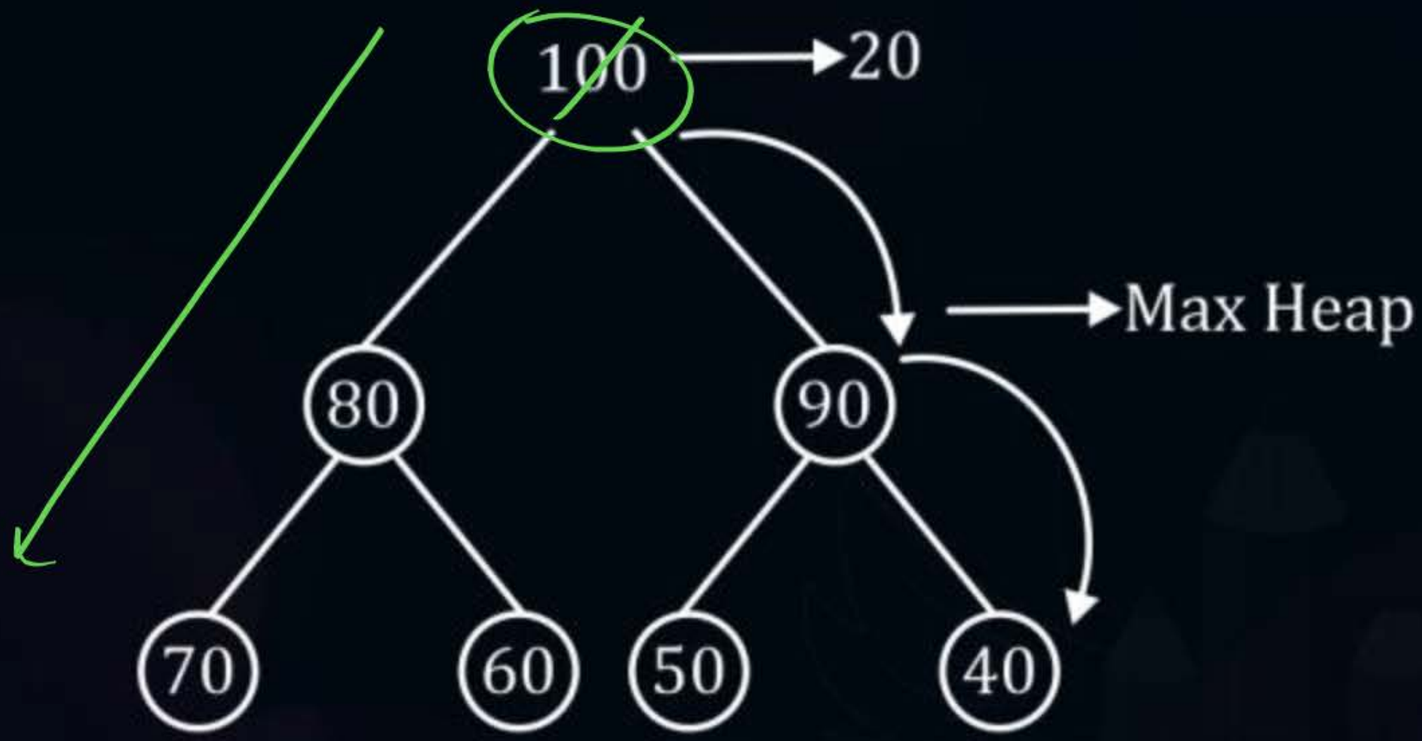
Increase /Decrease key operation: TC

Given a heap with ‘n’ elements the time complexity to perform the increasing key/decrease key operation is_____.



Topic : Heap Algorithm

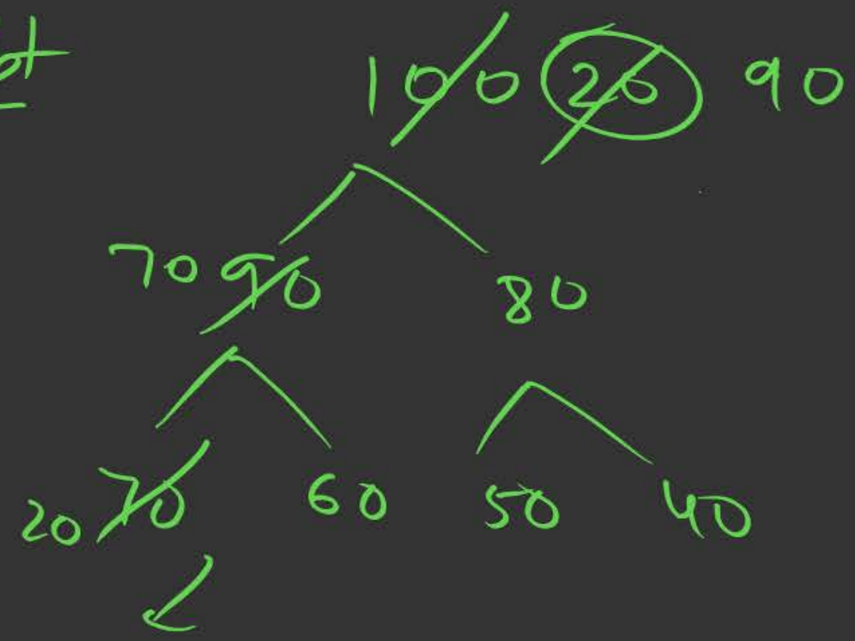


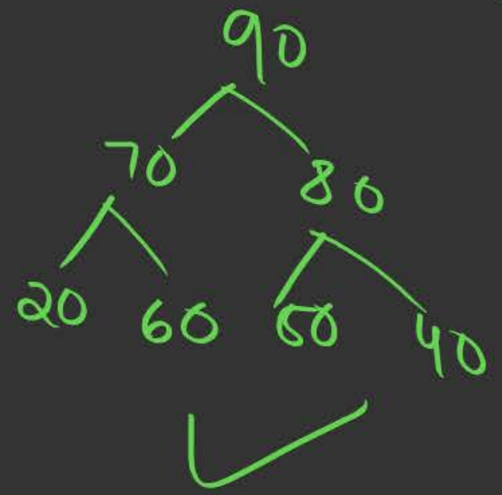


After decrease/ increase operation, we have to readjust to make it a valid heap again.

= O(logn)

Doubt







Topic : Heap Algorithm

V. Imp

Delete operation in a Heap

For a max Heap → Max elements

Min- Heap → min elements

} The root node is getting deleted.

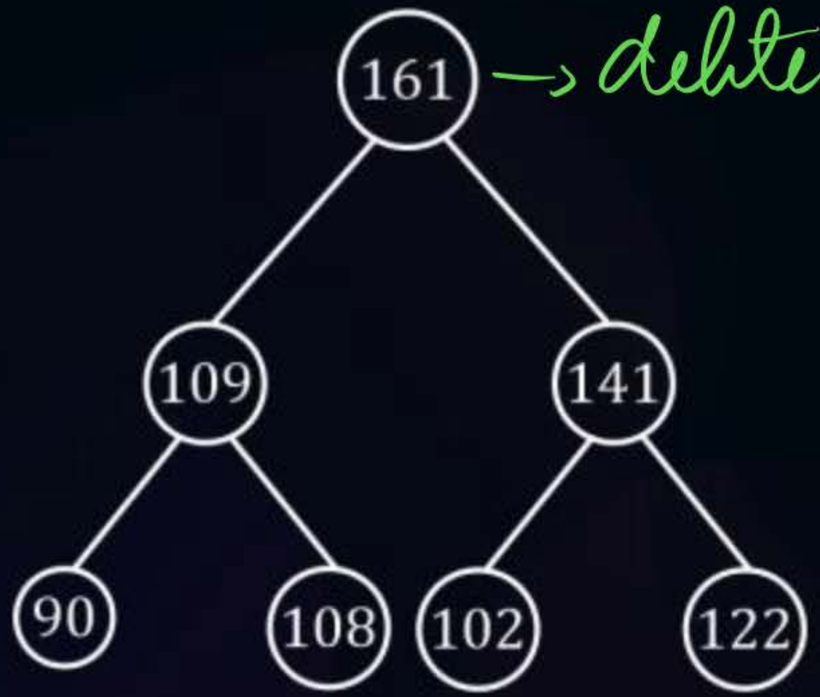
1. Swap Root with the last elements swap (A [i] , A [n])
2. Remove A [n]
3. Adjust A [i]





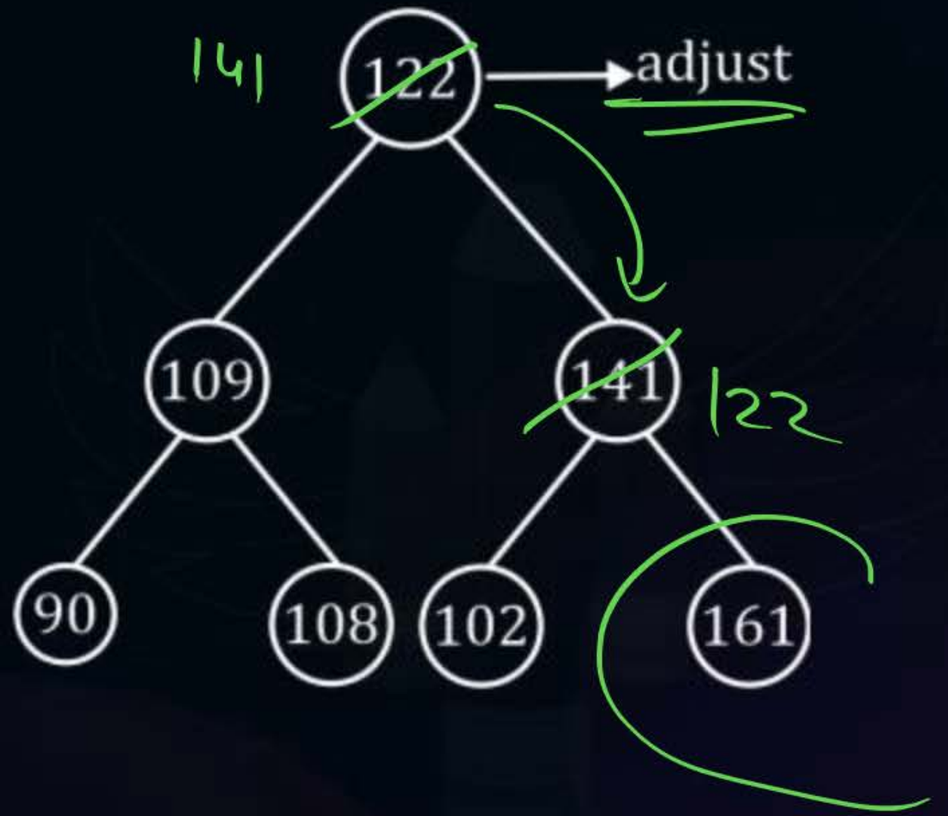
Topic : Heap Algorithm

E.g.2. Given Heap:



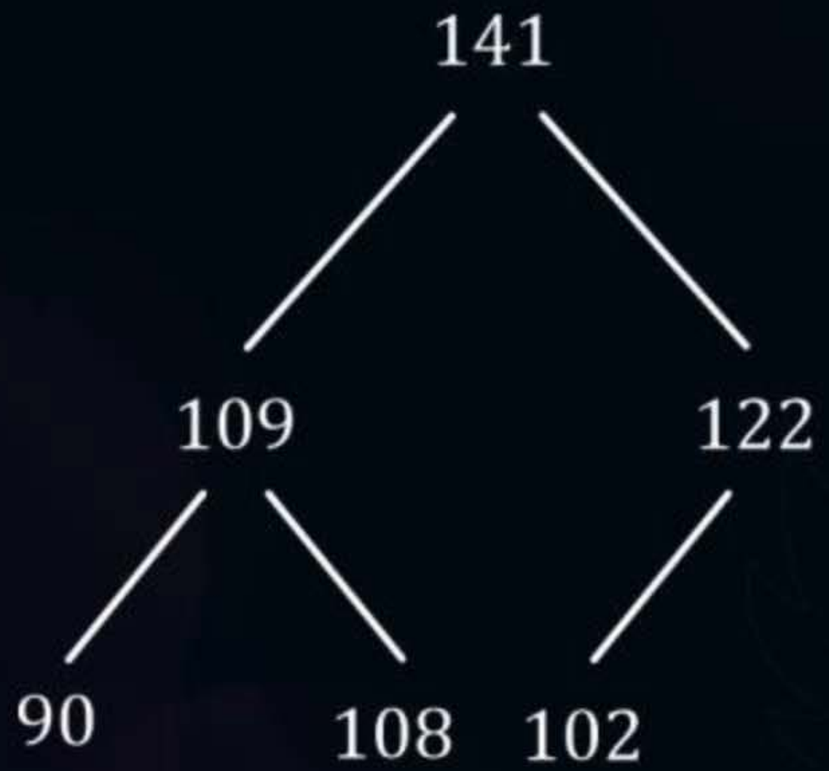
Step-1: ~~Swap (16, 1, 12, 2)~~

Swap(161, 122)





Topic : Heap Algorithm



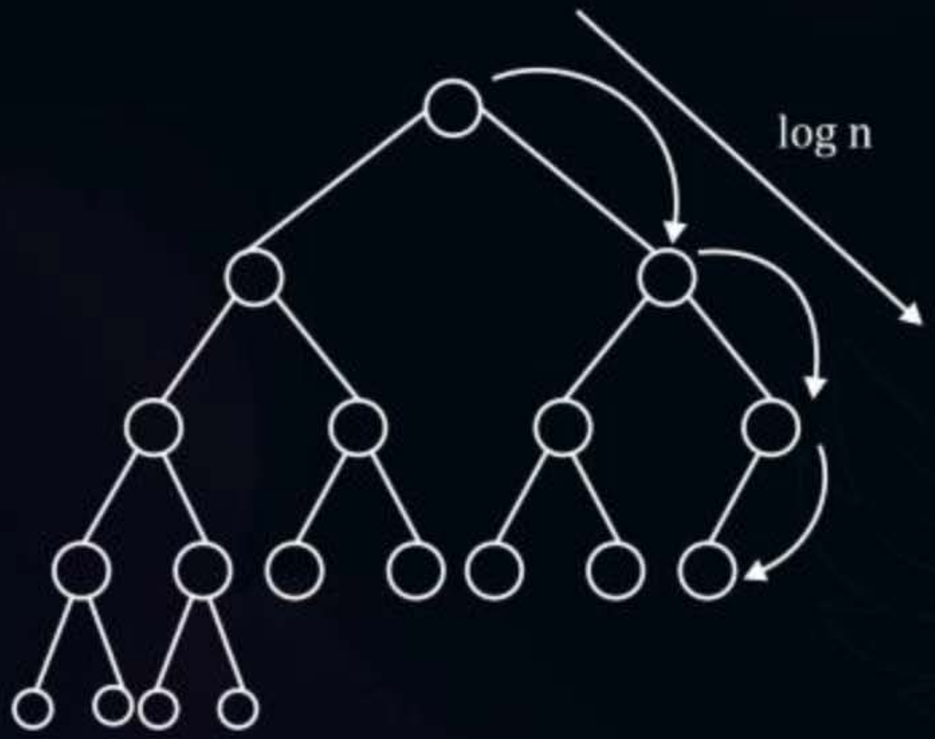
→ Final Heap after deleting 161.



Topic : Heap Algorithm

Time complexity of a Delete Operation :

O(log n)



Delete → O(log n)

Ater removing you have to adjust the new root to make sure its back a valid Heap.

Questions



Topic : Heap Algorithm

#Q. Which Array Representation is a valid Binary Max-Heap

100%

B

A <25, 12, 16, 13, 10, 8, 14>

B <25, 14, 16, 13, 10, 8, 12 >

C <25, 14, 13, 16, 10, 8, 12>

D <25, 14, 12, 13, 10, 8, 16>

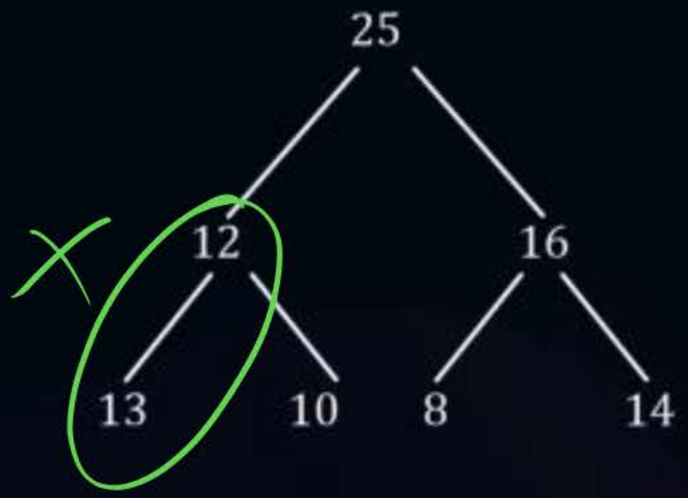
Part b) :- insert (7, 2, 10, 4)

0 Swaps

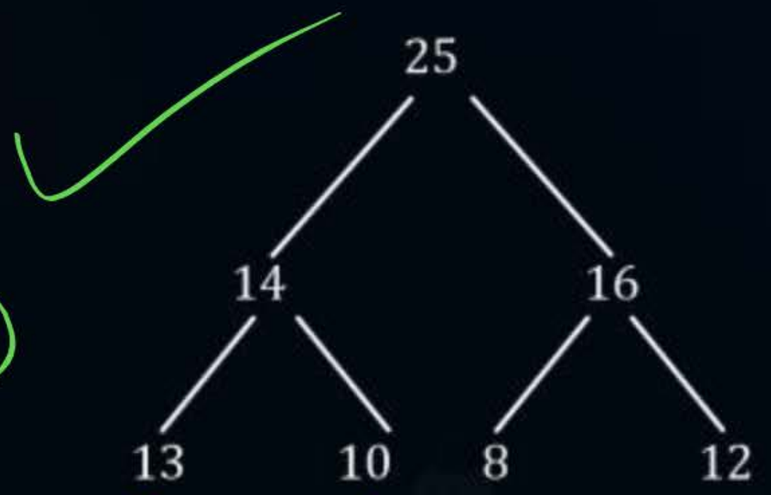


Topic : Heap Algorithm

(A)



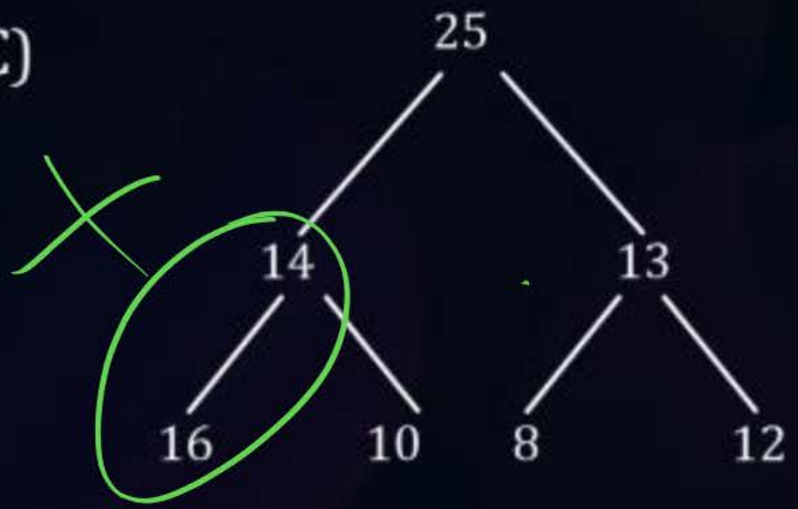
(B)



(7,2,10,4)



(C)





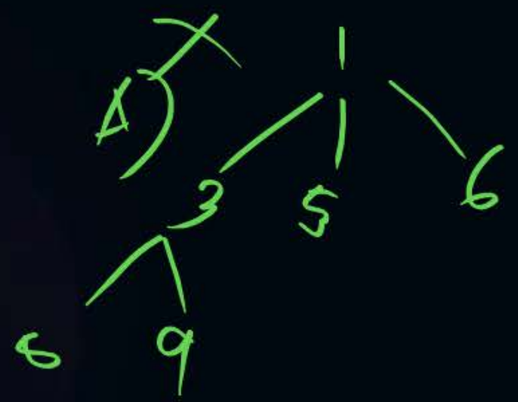
Topic : Heap Algorithm

#Q. Which one is ~~valid~~ are maximum Heap Array representation

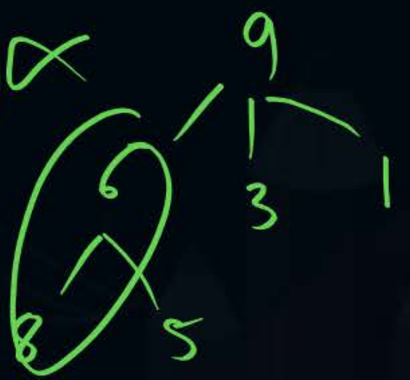
Valid ternary → atmost 3 child

Binary
Ternary

A) <1, 3, 5, 6, 8, 9> ✗



B) <9, 6, 3, 1, 8, 5> ✗



C) <9, 3, 6, 8, 5, 1> ✗



D) <9, 5, 6, 8, 3, 1> ✓



Heap Sort



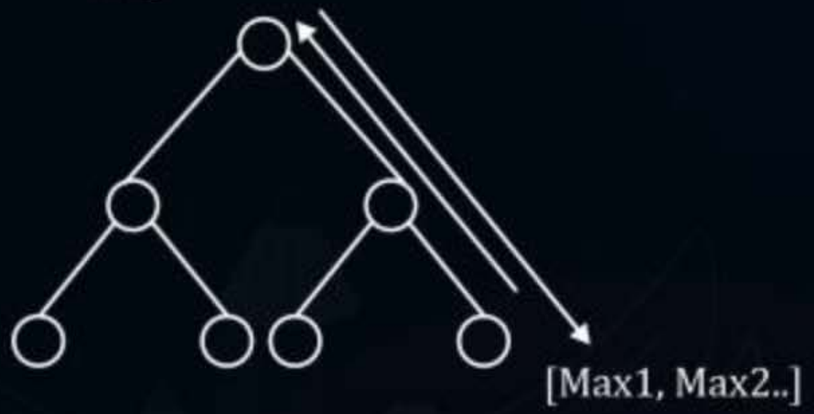
Topic : Heap Algorithm

```
procedure HEAPSORT (A, n)
// A(1 : n) contains n elements to be sorted.//
1.    call HEAPIFY (A, n)
2.    for i ← n to 2 by – 1 do
      {
        t ← A(i); A(i) ← A(1); A(i) ← t
        call ADJUST, (A, 1, i – 1)
      }
    repeat
    end HEAPSORT
```

n → 2

Topic : Heap Algorithm

Explanation:-  CBT → Heap

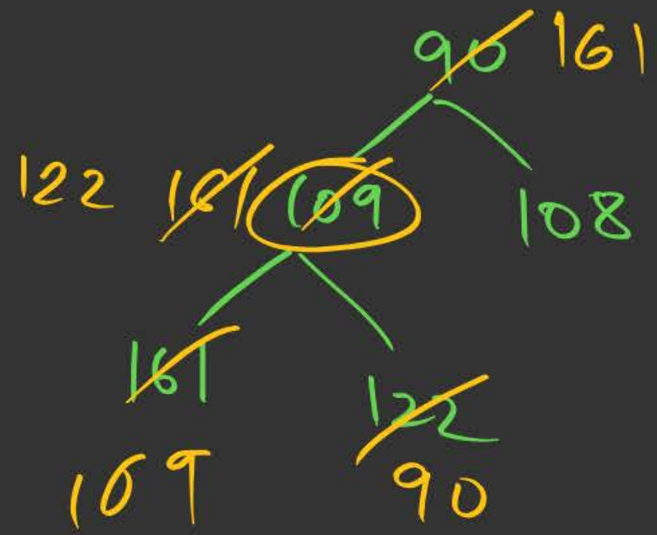


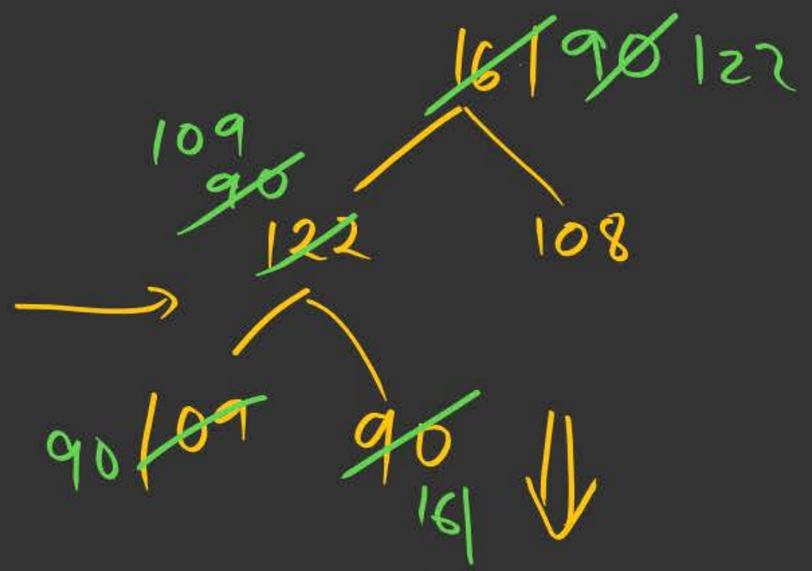
root→ always max element in the existing Heap.

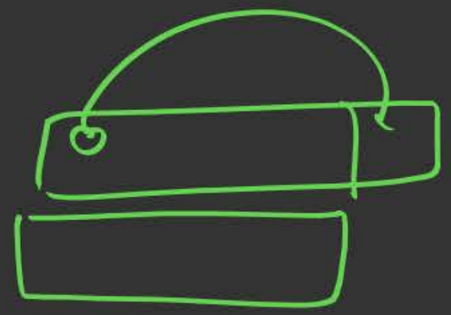
Logic:- Imp

Every time a root elements is deleted (max element) the delete operation will re-adjust the Heap such that the new maximum (max among remanning elements) becomes the root node (keep repeating this process.)

A = [90, 109, 108, 161, 122]







[161, 122, 108, 109, 90]

[122, 109, 108, 90, 161]



Topic : Heap Algorithm

HeapSort

Step by step iteration

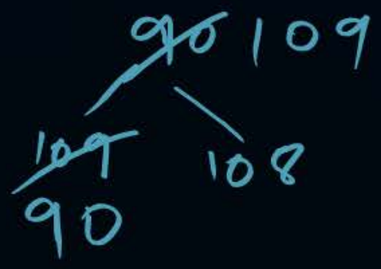
given A ⇒
Heapify →

	1	2	3	4	5
	90	109	108	161	122
	161	122	108	109	90
i=5	90	122	108	109	161
i=4	122	109	108	90	161
i=3	109	90	108	122	161
i=2	108	90	109	122	161
	90	108	109	122	161

i=n

90

→ Sorted in asc order

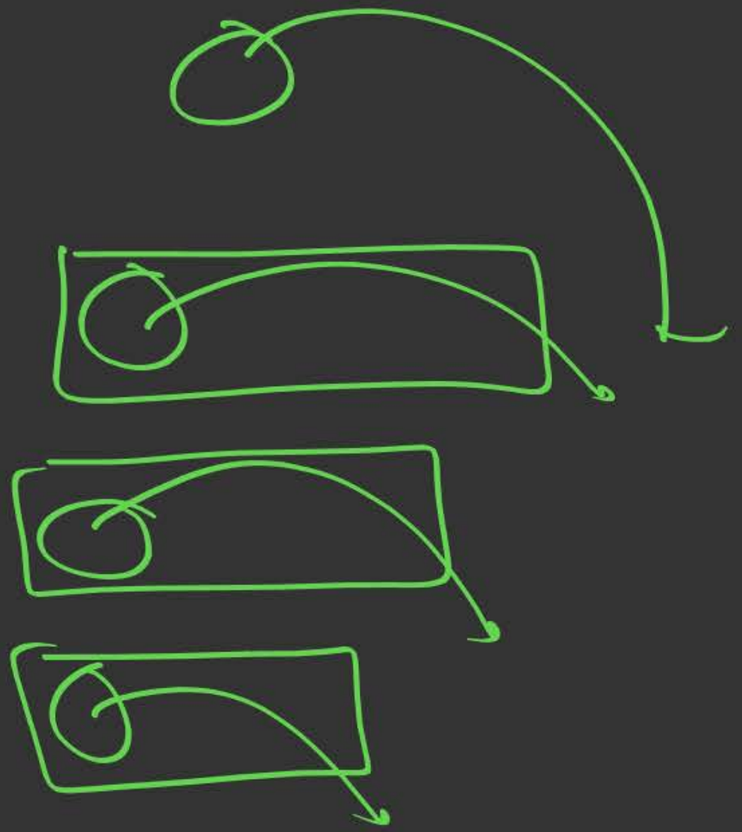






Heapify → Max Heap → asc

Heapify → Min Heap → desc





Topic : Heap Algorithm

Time complexity analysis:

Time complexity in every case $= \theta\ (n + n \log n)$

$= \theta\ (n + n \log n) = \Theta(n \log n)$

Space complexity $= O(1) \rightarrow$ Inplace

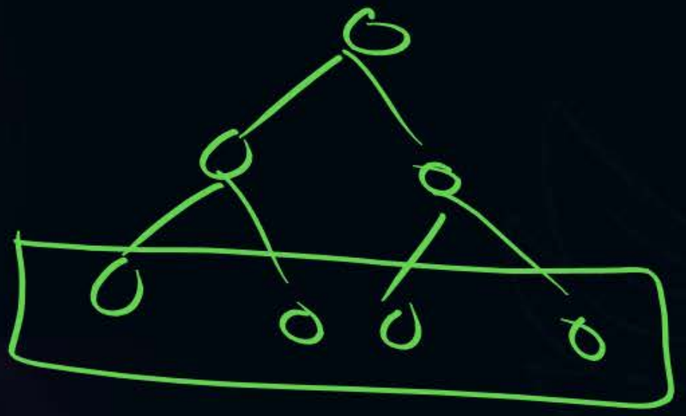
Not stable



Topic : Heap Algorithm

#Q. In a Binary Max-Heap with n elements, the smallest element can be found in time of _____.

59%.



A O(nlogn)

B O(n)

C $O(n^2)$

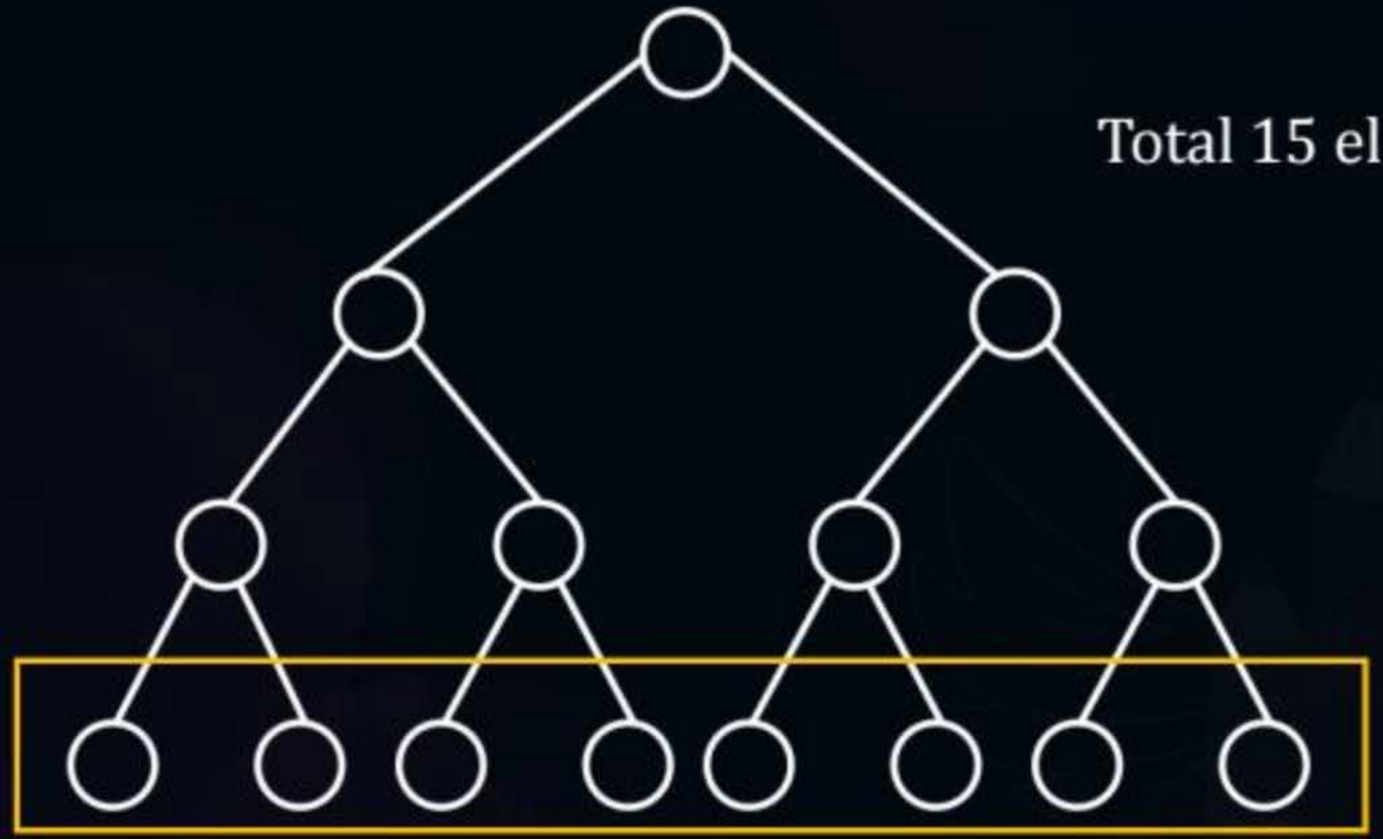
D O(1)



Topic : Heap Algorithm

Max- Heap

Approx



Total 15 elements → n

$n/2 \approx$

Smallest element always a leaf of max-Heap

$O(n/2) = O(n)$



Topic : Heap Algorithm

15	→	8
Total		last level

$n \rightarrow \simeq \left\lceil \frac{n}{2} \right\rceil$ at leaf level

Appr. Linear search once these n/2 elements → O(n)



Topic : Heap Algorithm

Appr. 2

H.W!



Given max-Heap perform heapify on this to convert it to a min-heap and then print the root node-(min element).

Heapify → O(n)



Topic : Heap Algorithm

HW2

#Q. Level order traversal of a binary max heap generates: <10, 8, 5, 3,2>. To this Heap Insert: <1 and 7>; What is the resultant level order Traversal

A 10, 8, 7, 3, 5, 21

B 10, 8, 7, 3, 2, 1, 5

C 10, 7, 8, 3, 5, 1, 2

D 10, 7, 8, 5, 3, 2, 1



2 mins Summary

- Topic
- Topic
- Topic
- Topic
- Topic

Heap

- Delete
- Heap Sort



THANK - YOU